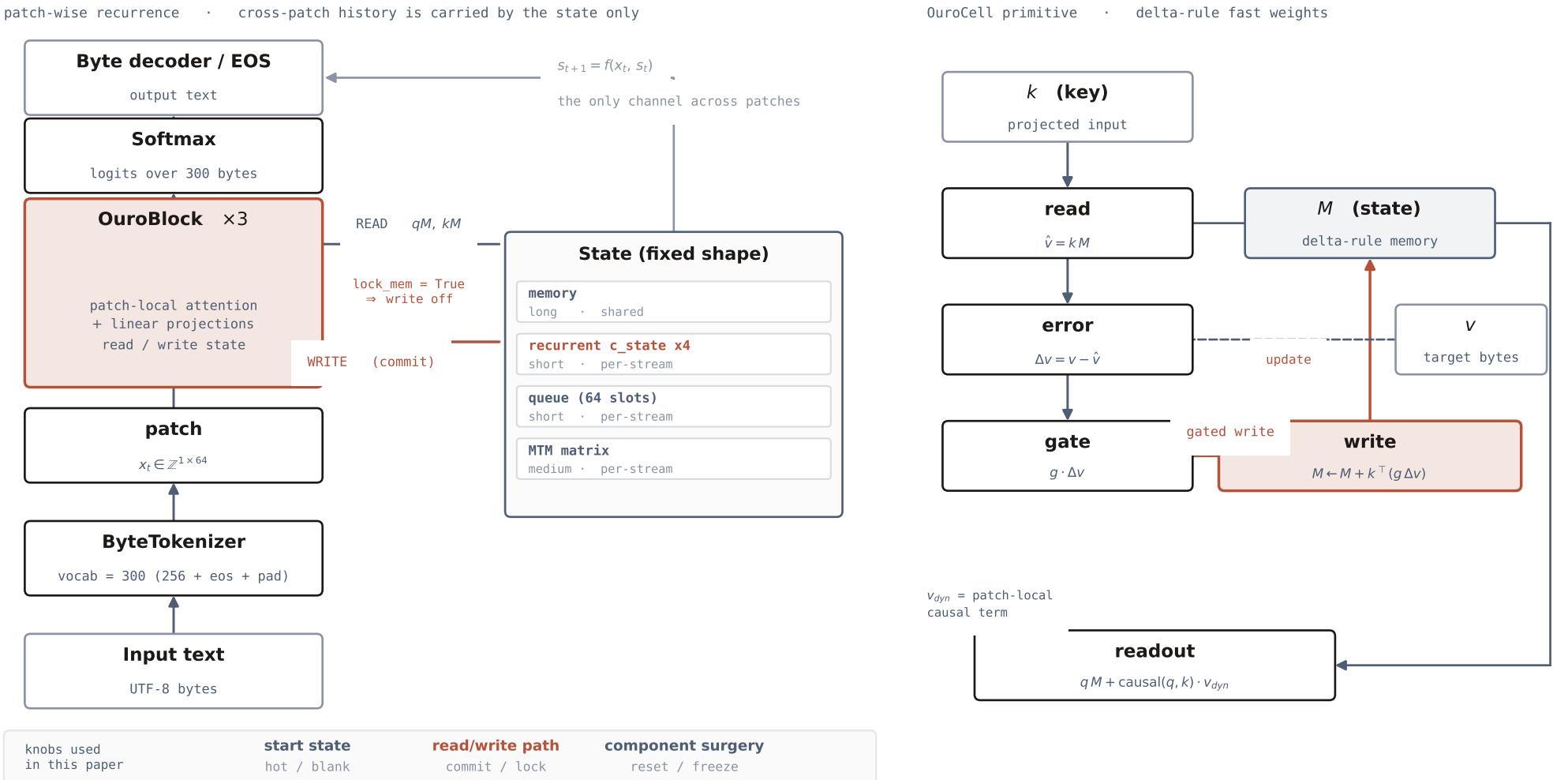

patch-wise recurrence · cross-patch history is carried by the state only
Byte decoder / EOS
output text
Softmax
logits over 300 bytes
OuroBlock ×3
patch-local attention
+ linear projections
read / write state
patch
$x_t \in \mathbb{Z}^{1\times 64}$
ByteTokenizer
vocab = 300 (256 + eos + pad)
Input text
UTF-8 bytes
$s_{t+1} = f(x_t, s_t)$
the only channel across patches
READ qM, kM
lock_mem = True
⇒ write off
WRITE (commit)
State (fixed shape)
memory
long · shared
recurrent c_state x4
short · per-stream
queue (64 slots)
short · per-stream
MTM matrix
medium · per-stream
knobs used
in this paper
start state
hot / blank
read/write path
commit / lock
component surgery
reset / freeze
OuroCell primitive · delta-rule fast weights
k (key)
projected input
read
$\hat{v} = kM$
M (state)
delta-rule memory
error
$\Delta v = v - \hat{v}$
update
v
target bytes
gate
$g \cdot \Delta v$
gated write
write
$M \leftarrow M + k^\top (g\,\Delta v)$
v_{dyn} = patch-local
causal term
readout
$qM + \text{causal}(q,k) \cdot v_{dyn}$